

Software-Based Virtual Motor Control Using PWM and PID

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1. Problem Statement

The objective of this project is to design and implement a virtual motor control system using PWM and PID control techniques. The system must generate appropriate control signals, simulate motor behavior accurately, and analyze the dynamic performance using simulation plots. This project is implemented in MATLAB Simulink and demonstrates the principles of control systems in practical applications.

2. System Overview

The overall system consists of a reference input, error calculation block, PID controller, PWM generator, DC motor model, and feedback loop. The reference signal defines the desired motor speed, and the controller adjusts the system to achieve this target. The signal flow follows: Reference → Error → PID → PWM → Motor → Feedback. Each component plays a critical role in ensuring stable and accurate control of the motor.

3. PWM Generation Logic

Pulse Width Modulation (PWM) is used to control the effective voltage applied to the motor. A high-frequency triangle wave is compared with a duty cycle signal generated by the PID controller. When the duty cycle is greater than the triangle signal, the output is HIGH (1), otherwise it is LOW (0). This results in a series of pulses whose width varies with the control signal. The average voltage applied to the motor is proportional to the duty cycle, given by: $V_{avg} = \text{Duty} \times V_{supply}$. This allows efficient and precise control of motor speed without continuous analog voltage variation.

4. PID Controller Design

The PID controller is used to minimize the error between the desired and actual motor speed. The control equation is: $u(t) = K_p \cdot e(t) + K_i \int e(t) dt + K_d \cdot de(t)/dt$. The proportional term reacts to present error and improves response speed. The integral term accumulates past error and ensures zero steady-state error. The derivative term predicts future error and reduces overshoot by adding damping. Proper tuning of these gains ensures a balance between speed, stability, and accuracy.

5. Motor Modeling

The DC motor is modeled using both electrical and mechanical equations. The electrical equation is: $V = L \cdot di/dt + R \cdot i + e_b$, where e_b is the back EMF. The mechanical equation is: $T = J \cdot d\omega/dt + B \cdot \omega + T_f$. Inertia (J) controls how quickly the motor accelerates, while viscous friction (B) provides damping. These parameters define the dynamic behavior of the motor and influence rise time, overshoot, and settling characteristics.

6. Closed-Loop Control

The system operates as a closed-loop control system where feedback is continuously used to correct the output. The actual motor speed is fed back and compared with the reference speed to produce an error signal. The PID controller processes this error and adjusts the duty cycle accordingly. This ensures that the system automatically corrects deviations and maintains the desired speed under varying conditions.

7. Performance Evaluation

The system performance is evaluated using simulation plots. The motor reaches the desired speed with a slight overshoot and then settles to the steady-state value. The rise time is approximately 1 to 1.5 seconds, and the settling time is around 4 to 5 seconds. The overshoot is approximately 10–12%, which is acceptable for a well-tuned system. The steady-state error is nearly zero, indicating accurate tracking of the reference speed.

8. Conclusion

The project successfully demonstrates the implementation of a PWM-based motor control system using a PID controller. The system achieves stable and accurate speed control with minimal steady-state error. PWM provides efficient voltage control, while the PID controller ensures dynamic performance. The results validate the effectiveness of closed-loop control systems in real-world motor applications.

9. Plots and Analysis

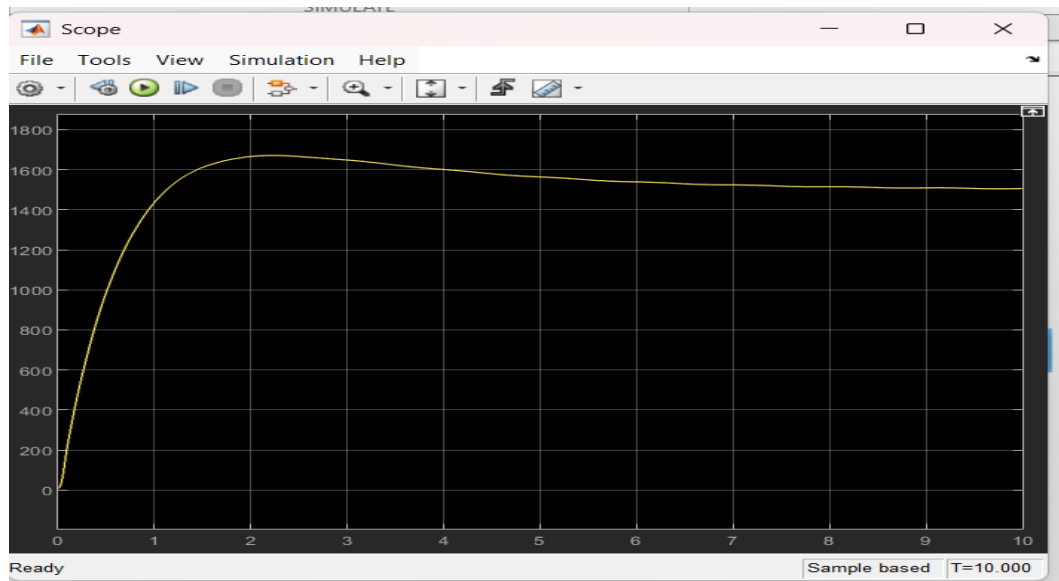


Figure 1: Speed Response - The motor speed rises rapidly, shows a small overshoot, and settles at the desired value of 1500 RPM, demonstrating stable closed-loop control.

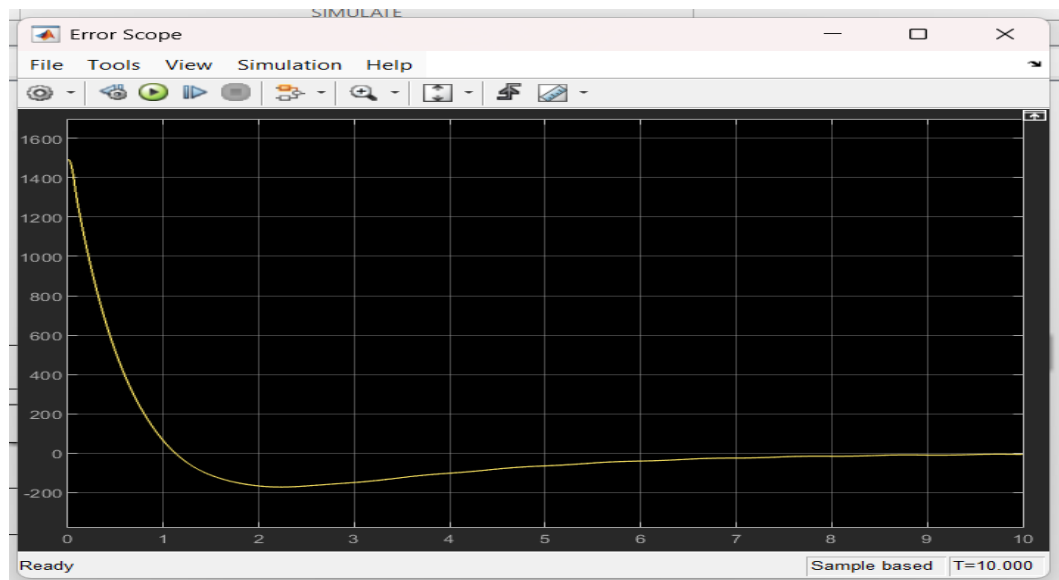


Figure 2: Error Response - The error starts high and gradually reduces to zero, indicating that the PID controller successfully minimizes the difference between reference and actual speed.

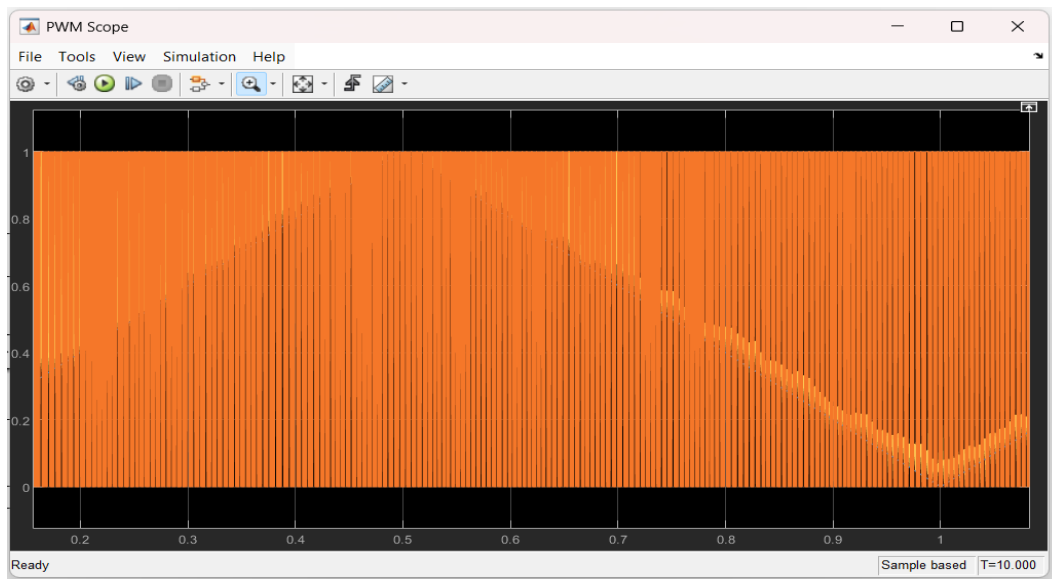


Figure 3: PWM Waveform - The PWM signal switches between 0 and 1, with pulse width varying according to the duty cycle, showing how the controller regulates motor voltage.